

NATIONAL PUBLIC SCHOOL BANGLORE



Project Report

BANKINATOR400

12th Grade Computer Science Project



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ABOUT THE PROJECT

We created a Python-based Virtual Bank Machine to replicate the basic functioning of a real banking system in a simple and user-friendly way.

The system allows users to create bank accounts and log in securely using passwords chosen by them.

After logging in, users can check their account balance, carry out credit and debit transactions, and view a complete history of their transactions. Each transaction is saved with important details such as the date, time, transaction type, and amount, helping users keep track of their financial activity.

A bill is generated and printed after every successful transaction. This bill acts as a receipt and displays the transaction details along with the updated account balance, similar to what users experience at an ATM or bank counter.

In addition to user features, we included an administrative module that provides higher-level control over the system. Administrators can view all existing accounts in the bank, approve or reject new account creation requests, and delete accounts when required. For better organization and improved security, user passwords are stored in a separate table instead of being mixed with account data.

MODULES USED

These are the modules we used -

1. `create_account()` - Misc - creates an account and stores it in a table aptly named `acctobeapproved`, with passwords stored in `accpwdtba`
2. `approve_account()` - Admin - approves an account whose account number is accepted as input, and moves the record from `acctobeapproved` to `bank_accounts`, and passwords from `accpwdtba` to `accpwd`, and creates a custom user table with the name as `'acc'+account number`, eg - `acc1000000001`, to store transaction details
3. `check_balance(acc)` - User - checks the balance of a user in `bank_accounts`
4. `transaction(acc)` - User - makes a transaction (credit/debit), changes balance in `bank_accounts`, logs a transaction into account table, returns a bill with new balance, transaction amount+type, time and date of transaction
5. `delete_account()` - Admin - deletes the account in `bank_accounts`, deletes password from `accpwd`, and deletes account table from database
6. `check_transaction()` - User - retrieves the last 10 account transactions made in the form of a table
7. `display_all()` - Admin - displays all records in `bank_accounts`
8. `user_login_acc(acc)` - Misc - allows users to login to access features offered by Bankinator400

SCREENSHOTS

```
===== RESTART: D:\Anirudh K and Athreya S I2A Project\Bankinator400.py =====
Connected Successfully
-----
Enter -
1 to login as user
2 to login as admin
3 to create new account :
```

User login fail -

```
1

Enter account number : 1000000001
Enter password : accpwd1
Incorrect PIN...
Press enter to exit or any key to stay :
```

User login successful, loading screen

```
1

Enter account number : 1000000001
Enter password : user1PWD
-----
Welcome to

BANKINATOR400

-----
Enter-
1 to check bank account status,
2 to make a transaction
3 to check recent transactions :
```

Check bank account status :

```
1

+-----+-----+-----+
| Account_no | Name   | Current Balance |
+=====+=====+=====+
| 1000000001 | user1  |          4000   |
+-----+-----+-----+
```

Check_transaction(acc)-

```
3

Enter account number : 1000000001

+-----+-----+-----+
| Date and time of transaction | Transaction Type | Transaction Amount |
+-----+-----+-----+
| 2025-11-03 10:19:00          | credit          |          900       |
+-----+-----+-----+
| 2025-11-03 10:13:57          | credit          |         999999     |
+-----+-----+-----+
```

Transaction (credit) –

```
2

Enter transaction type : credit
Enter transaction amount : 1000

-----

Bankinator 400

Bill

Date and time : 2025-12-10 13:51:29.927115
Account Details :-
Account Number : 1000000001
Name : user1

Transaction Type : Credit
Transaction Amount : 1000

-----
```

After transaction –

```
3
Enter account number : 1000000001
+-----+-----+-----+
| Date and time of transaction | Transaction Type | Transaction Amount |
+-----+-----+-----+
| 2025-12-18 13:51:30          | credit           | 1000                |
+-----+-----+-----+
| 2025-11-03 10:19:00          | credit           | 900                  |
+-----+-----+-----+
| 2025-11-03 10:13:57          | credit           | 999999              |
+-----+-----+-----+
```

```
1
+-----+-----+-----+
| Account_no | Name | Current Balance |
+-----+-----+-----+
| 1000000001 | user1 | 5000             |
+-----+-----+-----+
```

Transaction debit –

```
2
Enter transaction type : debit
Enter transaction amount : 3999

-----

Bankinator 400 Bill
Date and time : 2025-12-18 13:53:38.876421
Account Details :-
Account Number : 1000000001
Name : user1

Transaction Type : Debit
Transaction Amount : 3999

-----
```

After transaction –

1

Account_no	Name	Current Balance
10000000001	user1	1001

3

Enter account number : 10000000001

Date and time of transaction	Transaction Type	Transaction Amount
2025-12-18 13:53:39	debit	3999
2025-12-18 13:51:30	credit	1000
2025-11-03 10:19:00	credit	900
2025-11-03 10:13:57	credit	999999

Transaction fail because of too less resulting bank account balance –

2

Enter transaction type : debit
Enter transaction amount : 400
Current balance is not enough...

Logout as user –

```
Enter 'Logout' to leave or any key to stay : logout  
Press enter to exit or any key to stay : a  
-----
```

```
Enter -  
1 to login as user  
2 to login as admin  
3 to create new account :
```

Login fail as admin –

```
Enter 'Logout' to leave or any key to stay : logout  
Press enter to exit or any key to stay : a  
-----
```

```
Enter -  
1 to login as user  
2 to login as admin  
3 to create new account :
```

Admin login success –

```
Enter -  
1 to login as user  
2 to login as admin  
3 to create new account :
```

```
2
```

```
Enter admin password : adminpass123  
-----
```

```
Welcome to BANKINATOR4000  
-----
```

```
Enter-  
1 to display all accounts,  
2 to approve pending accounts,  
3 to delete account :
```


Option 1-

1

Account_no	Name	Current Balance
1000000001	user1	1001
1000000006	Tanushri	1001

Option 2 – account approval –

2

Account Number	Name	Current Balance
1000000002	user2	89999
1000000003	user3	79999
1000000004	user7	39999

Account approved (enter account number): 1000000002

Press enter to exit or any key to stay : |

Option 3 –

```
2

+-----+-----+-----+
| Account Number | Name   | Current Balance |
+-----+-----+-----+
| 1000000002    | user2  | 89999           |
+-----+-----+-----+
| 1000000003    | user3  | 79999           |
+-----+-----+-----+
| 1000000004    | user7  | 39999           |
+-----+-----+-----+

Account approved (enter account number): 1000000002

-----
Press enter to exit or any key to stay : |
```

After approving/deleting –

```
1

+-----+-----+-----+
| Account_no | Name   | Current Balance |
+-----+-----+-----+
| 1000000001 | user1  | 1001            |
+-----+-----+-----+
| 1000000002 | user2  | 89999           |
+-----+-----+-----+
```

Creating new account –

```
-----
Creating account...
-----

Enter your name : user5
Enter password (8 min ; 30 max):
user5PWD
Password is valid.

Enter deposit ammount : 9000

-----

Account number : 1000000006
Registered details of account number : 1000000006
+-----+-----+-----+
| Account number | Name   | Current balance |
+-----+-----+-----+
|      1000000006 | user5  |           9000 |
+-----+-----+-----+

Account pending approval...

Press enter to exit or any key to stay : |
```

Password creation invalid –

```
Enter your name : user10
Enter password (8 min ; 30 max):
asdff
Password is invalid.
Password must meet the following criteria :
1. Must be between 8-30 characters long.
2. Must be a mix of letters and numbers.

Please enter a valid password :
|
```

Deposit amount too low –

```
Please enter a valid password :  
user10PWD  
Password is valid.  
  
Enter deposit ammount : 900  
Error: deposit entered is too low.  
Minimum balance is 1000 Rs.  
Enter deposit amount :
```

Corrected-

```
Enter deposit ammount : 900  
Error: deposit entered is too low.  
Minimum balance is 1000 Rs.  
Enter deposit amount : 1000  
  
-----  
Account number : 1000000007  
Registered details of account number : 1000000007  
+-----+-----+-----+  
| Account number | Name   | Current balance |  
+-----+-----+-----+  
| 1000000007    | user10 | 1000            |  
+-----+-----+-----+  
  
Account pending approval...
```

Tables created in database –

```
mysql> show tables;
+-----+
| Tables_in_project |
+-----+
| acc10000000001    |
| acc10000000002    |
| acc10000000005    |
| accpwd             |
| accpwdtba          |
| acctobeapproved    |
| bank_accounts      |
+-----+
7 rows in set (0.00 sec)
```

Account table desc –

```
mysql> describe acc10000000001
-> ;
+-----+-----+-----+-----+-----+-----+
| Field                | Type         | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| Date                 | datetime    | NO   | PRI | NULL    |       |
| Transaction_Type     | varchar(20) | YES  |     | NULL    |       |
| Transaction_Amount   | int         | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
```

Bank_accounts table desc-

Field	Type	Null	Key	Default	Extra
Account_No	int	NO	PRI	NULL	
Name	varchar(30)	YES		NULL	
Current_Balance	int	YES		NULL	

3 rows in set (0.00 sec)

Acctobeapproved desc –

```
mysql> desc acctobeapproved;
```

Field	Type	Null	Key	Default	Extra
Account_No	int	NO	PRI	NULL	
Name	varchar(30)	YES		NULL	
Current_Balance	int	YES		NULL	

3 rows in set (0.00 sec)

Desc accpwdtba and accpwd –

```
mysql> desc accpwd;
```

Field	Type	Null	Key	Default	Extra
Account_No	int	NO	PRI	NULL	
PIN	varchar(30)	YES		NULL	

2 rows in set (0.00 sec)

```
mysql> desc accpwdtba  
-> ;
```

Field	Type	Null	Key	Default	Extra
Account_No	int	NO	PRI	NULL	
PIN	varchar(30)	YES		NULL	

2 rows in set (0.00 sec)

FLOW CHART

